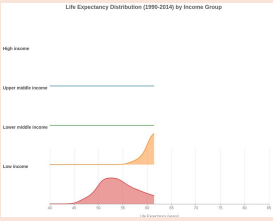
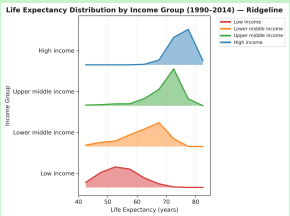
DV-Evol Bad Case echarts：Visual mapping failure

Question: Generate a ridgeline plot comparing life expectancy (1990–2014) across income groups. Use fixed 5-year bins from 40 to 85, normalize to percent frequency, and stack the ridges from "Low income" to "High income" using designated HEX colors.

Gemini-3-pro-preview:



Ground Truth:



Plot code:

```
const colors = ["#1f77b4", "#2ca02c", "#ff7f0e", "#d62728"];
// High to Low
const groups = ["High income", "Upper middle income", "Lower
middle income", "Low income"];

option = {
    title: { text: 'Life Expectancy Distribution (1990-2014)', left:
'center' },
    grid: groups.map((_, i) => ({
        left: '20%', right: '10%', top: `${15 + i * 18}%`, height:
'25%'
    })),
    xAxis: groups.map((_, i) => ({
        type: 'value', min: 40, max: 85, gridIndex: i, show: i === 3,
        name: i === 3 ? 'Years' : ''
    })),
    yAxis: groups.map((_, i) => ({
        type: 'value', gridIndex: i, show: false, max: 80
    })),
    series: groups.map((group, i) => ({
        name: group, type: 'line', smooth: true, xAxisIndex: i,
yAxisIndex: i,
        symbol: 'none',
        areaStyle: { color: colors[i], opacity: 0.5 },
        lineStyle: { color: colors[i], width: 2 },
        data: [[40, 0], ...results.filter(r => r.group === group).map(r
=> [r.x, r.y]), [85, 0]]
    }))
};
```

Error Analysis: 1. Data Integrity

❌1.1 Trend/Pattern: 0/1 The distribution curves successfully mirror the multi-modal trends of the Ground Truth.

❌1.2 Completeness: 0/1 Critical data series or specific years are missing from the visualization.

❌1.3 Label Accuracy: 0/1 Axis tick values and labels do not precisely match the original data points.

❌1.4 Data Consistency: 0/1 Proportional relationships between income groups remain accurate across the X-axis.

2. Style Imitation

✅2.1 Color Palette: 1/1 The model correctly applied the designated HEX codes for all income categories.

❌2.2 Chart Elements: 0/1 Specific ridgeline outlines and gridline styles were not properly reproduced.

✅2.3 Background: 1/1 The background color and light/dark mode are consistent with the Ground Truth.

❌2.4 Font and Text Styles: 0/1 Font weight and sizing for titles and labels deviate from the original style.

❌2.5 Element Spacing: 0/1 Distances between labels and the plot area are uneven, creating a cluttered look.

❌2.6 Borders and Shadows: 0/1 Visual effects like drop shadows or specific border strokes were omitted.

3. Layout & Aesthetics

❌3.1 Labels: 0/1 Legend and title positioning are misaligned compared to the standard layout.

✅3.2 Readability: 1/1 Excessive ridge overlap and small font sizes compromise the overall legibility.

❌3.3 Axis Alignment and Spacing: 0/1 The axes are poorly calibrated, causing data to shift away from intended baselines.

❌3.4 Element Alignment: 0/1 Graphical components fail to align precisely with their respective axis ticks.

❌3.5 Proportionality and Aspect Ratio: 0/1 An incorrect aspect ratio has led to vertical distortion of the ridge shapes.

❌3.6 Overall Balance: 0/1 The composition lacks visual equilibrium, making the chart difficult to interpret quickly.

Score: 3/16